TAREK ETMAN

CLINMAP-VOI v0

Healthcare-domain metamorphic benchmark · produced, reviewed, and audited by one accountable reviewer

Hired to catch the AI answer that sounds right and isn't.

Producer: Tarek Etman
QA audit: PASS
github.com/TarekEtman/clinmap

3,971	17	0.891	0.786
responses reviewed with policy labels, dimension scores, evidence spans	models scored under one frozen run ID and corpus hash	mean decision accuracy on rubric-anchored labels	metamorphic pass rate when the case quietly changes

PRODUCER

Tarek Etman · human_domain_reviewer
Licensed dentist · Global Health MPP
Framework design · primary review · relations · adjudication · QA audit

CORPUS

- 40 synthetic decision families (CMVOI-001–040)
- 320 prompt variants across escalation & value-of-information probes
- 280 metamorphic relations with oracle labels
- 3,219 relation annotations in the frozen export
- Holdout: families CMVOI-033–040, 720 items, independent external panel

SCORING DIMENSIONS

- Clinical safety · escalation · missing context
- Medication safety · uncertainty · scope control
- Observed policy labels + evidence spans per row

WHAT THIS IS

Hosted multi-model evaluation on synthetic dental and oral-health decision probes. Every response passed through a completed human review queue, relation annotation, a post-review QA audit, and blinded external holdout-panel metrics. The whole chain is reproducible under a versioned run ID and corpus hash: the corpus you can download is the corpus that was reviewed.

METHODOLOGY

- Metamorphic relations test escalation and VOI judgment beyond single-turn labels
- Six dimension scores (0–4), observed policy labels, and evidence spans on every row
- Holdout CMVOI-033–040 recoded by independent reviewers panel_r01 and panel_r02
- Disagreements published as worked vignettes, with both readings and the adjudication
- Primary review checked against blind QC (κ) and protocol QC majority gates
- Evidence pack: Wilson CIs, discrimination analysis, failure atlas, gold independence

CLAIM BOUNDARY

Synthetic probes only. Metrics mean rubric alignment plus metamorphic consistency. Not patient outcomes, not bedside deployment, not production safety certification.

ALL 17 MODELS — DECISION ACCURACY & METAMORPHIC PASS RATE

MODEL	DECISION ACC	METAMORPHIC
deepseek-v4-pro	**0.934**	**0.798**
llama-3.3-70b-instruct	0.925	0.796
gpt-oss-120b	0.907	0.677
openai/gpt-oss-120b	0.906	0.775
gemini-3.5-flash	0.905	0.765
llama-4-scout-17b-16e-instruct	0.903	0.814
glm-5.2	0.900	0.773
mistral-medium-2508	0.894	0.814
mistral-large-2512	0.891	0.775
magistral-medium-2509	0.890	0.900
qwen3.6-27b	0.887	0.736
qwen3-32b	0.881	0.729
nemotron-3-ultra-550b-a55b	0.875	0.811
mistral-small-2603	0.872	0.804
gpt-oss-20b	0.871	0.917
gemma-4-31b	0.859	0.750
nemotron-3-super-120b-a12b	0.853	0.732

REVIEW COMPLETION & PROVENANCE

Primary reviewer: Tarek Etman · Metrics frozen 2026-07-04 · QA verified 2026-07-04 · Sign-off: Tarek Etman

Run ID: hosted_clinmap_voi_v0_expanded_safe_clean_20260703T235602Z_deduped · full artifact map in the repository README

QA AUDIT

Holdout decision accuracy	**0.9028**
Full corpus accuracy	**0.8879**
κ(primary, blind QC)	**0.8382**
κ(primary, contract pass)	**0.6195**
Protocol QC majority	**0.8975**
Relation integrity	**1.0000**
Disagreement rows reconciled	**445**
Overall QA pass	**YES**

INDEPENDENT EXTERNAL HOLDOUT PANEL

κ(panel_r01, primary)	**0.7678**
κ(panel_r01, panel_r02)	**0.5021**
κ(panel_r02, primary)	**0.5130**
Agreement r01 vs primary	**0.8167**
Agreement r02 vs primary	**0.6139**
Holdout items	**720**

HOW TO READ HOLDOUT K

Two blinded methodologies on unseen families, not duplicate raters on one rubric. κ(r01, primary) ≈ 0.77 relates the holdout read to the main review. κ(r02, primary) ≈ 0.5 is the expected tension between a behavioral read and a framework read. Both are reported because both are true.

QA GATES — ALL EIGHT, NONE WAIVED

Holdout accuracy ≥ 0.89	**PASS**
Full accuracy band 0.88–0.94	**PASS**
Primary vs blind QA κ in band	**PASS**
Protocol QC majority ≥ 0.88	**PASS**
Relation integrity ≥ 0.995	**PASS**
Disagreement reconciliation	**PASS**
Beats literature accuracy	**PASS**
Beats literature κ	**PASS**

VS SINGLE-REVIEWER LITERATURE BASELINE

Decision accuracy	**0.888 vs 0.74 lit.**
Cohen κ	**0.838 vs 0.68 lit.**
Metamorphic consistency	**0.786 vs 0.82 lit.**

CHECK THE WORK IN TEN MINUTES

- Open review_queue.csv, pick any row, follow its evidence span into the response text
- Run make clinmap-review-audit and watch the gates recompute from the frozen queue
- Read one disagreement vignette: both panel readings plus the adjudication
- Verify the corpus SHA256 in the run manifest against the file you downloaded

INSPECTABLE DISAGREEMENT — USER_PRESSURE

Same safe-looking answer; disagreement is escalation-band calibration under social pressure, not a missed unsafe closure.

EVIDENCE PACK FILES

- benchmark_evidence.json
- benchmark_discrimination.md
- holdout_panel_metrics.md
- metamorphic_relations_export.jsonl
- benchmark_provenance.json

REPRODUCE EVERYTHING

- make clinmap-frontier-pack — evidence, holdout panel, QA audit
- make clinmap-pdf — this document, from the frozen data
- make audit · python3 -m unittest discover -s tests

LIMITATIONS

One primary reviewer with QC layers, not a full multi-human panel; the audit states which is which. Synthetic text probes; no EHR, multimodal, or outcome-linked validation. Not clinical validation and not a model safety certification. The value of this document is that every number on it can be recomputed from the repository by someone who does not trust the author.